

Harshit Kumar Singh

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[LinkedIn](#) — [GitHub](#)

Technical Summary

Learning Python (Pandas, NumPy, Matplotlib, Scikit-learn, TensorFlow), Data Analysis, Machine Learning, Data Visualization, SQL, Optimization, Mathematical Modeling, and Research-oriented problem solving. Experienced in building analytical pipelines, designing surveys, processing datasets, and deploying interactive dashboards.

Objective

Motivated B.Tech student in IT and Mathematics seeking opportunities in **data science, machine learning, and data analytics**. Strong foundation in mathematical modeling, programming, and applied statistics. Eager to work on data-driven projects that contribute to real-world problem solving.

Education

Cluster Innovation Centre, University of Delhi

Expected - 2027

Bachelor of Technology in IT and Mathematics

Grade: 8.77 GPA

Relevant Courses: Discrete Mathematics, Calculus, Data Structures and Algorithms, Database Management, Probability and Statistics, Linear programming, Data Science.

Govt Boys Senior Secondary School No.1, Badarpur

April - 2023

Senior Secondary Education

Grade: 89.8%

Subjects: Hindi, English, Mathematics, Physics, Chemistry

Projects

Match-Making Optimization using Linear Programming	November 2025
- Applied Linear Programming (Hungarian Method) in Python to compute optimal pairings based on personality, interests, and lifestyle traits. - Designed the complete questionnaire and converted qualitative responses into a structured numerical compatibility matrix. Tools: Python (NumPy, Pandas)	
Himachal Pradesh District Road Safety Index & Forecasting <i>Data Science Project</i>	November 2025
- Built a Road Safety Index using accident severity and landslide exposure data. - Ranked all districts using normalized composite scoring. - Developed ML models to forecast district-level safety trends and identify risk-prone areas. Tools: Python, Pandas, NumPy, Scikit-learn, Matplotlib	
Predictive Analytics for TB Treatment Outcomes <i>Machine Learning and Neural Network Project</i>	October, 2025

- Developed ML and neural network models to predict treatment outcomes using patient-level clinical and facility data. - Identified key predictors through feature importance analysis, improving interpretability of high-risk patient profiles. - Evaluated multiple models (Logistic Regression, Random Forest, XGBoost, SVM, Neural Networks) to benchmark performance and reliability. Tools: Python, Pandas, Scikit-learn, TensorFlow:	
Time-Dependent Population Growth Model	December 2024
- Developed a mathematical model incorporating a time-dependent growth rate to analyze population dynamics. - Applied the model to various countries to study demographic transitions and compare results with traditional methods. Tools: Wolfram Mathematica, Ordinary Differential Equations	
Developed a copula-based financial risk modeling tool in Python	April 2025
- Simulating joint stock return dependencies using Gaussian, Student's t, and Vine Copulas. - Implemented marginal fitting, tail dependence analysis, CVaR, QQ plots, and Frobenius distance metrics. - Built a web interface to visualize synthetic data and extreme event simulations.	
Autonomous Drone Path Simulation (First-Principles Approach)	March 2026
- Built a UAV pathfinding simulation engine from scratch using numerical methods, without relying on external physics engines. - Modeled kinematics, collision detection, and trajectory optimization based on first-principle equations. - Implemented efficient path planning under dynamic constraints and obstacle environment	
Biometric-Based Attendance Management System (Android)	
- Designed and developed an end-to-end attendance system where instructors initiate session-based attendance and students authenticate using device biometrics. - Built a centralized database of students, enabling class-wise and subject-wise attendance tracking and reporting. - Implemented secure validation to prevent proxy attendance, with real-time synchronization between student devices and instructor dashboard.	

Skills

Programming	Python (Pandas, NumPy, Matplotlib, Scikit-learn, TensorFlow), SQL, C
Data Skills	EDA, Feature Engineering, Data Cleaning, Visualization, ML Modeling
Tools	Colab Notebook, Git, Excel, MySQL
Math & Stats Tools	MATLAB, R, Wolfram Mathematica
Other Skills	Optimization, Algorithmic Thinking, Survey Design, Model Interpretation
Soft Skills	Communication, Problem-Solving, Adaptability
Languages	Hindi (Native), English (Bilingual)

Technical Certificates

Programming with Gen AI	Oct 2025
Scientific Computing using Python (Hindi)	May 2025
The Joy of Computing using Python — Top 5%	May 2025
Design and Analysis of Algorithms	Apr 2025

Additional Certificates

Certificate	Date
Soft Skills (NPTEL) — Score 88	2025
English Language for Competitive Exams — Score 87	2025
Introduction to Professional Scientific Communication — Top 1%	Apr 2025
Psychology of Stress, Health & Well-Being — Top 5%	May 2025
Constitutional Law & Public Administration in India — Top 5%	May 2025

Experience

Volunteer, Unnat Bharat Abhiyan

Oct 2024 – May 2025

Contributed to bridging educational gaps for young students in underprivileged communities. Assisted in setting up a small library that operates weekly, fostering a reading culture among students.

Intern at Udhmodya Foundation – Startup Incubator, University of Delhi; contributed to social media management, website updates, report preparation, and sheet designing.

May 2025 – July 2025